EXPERIMENT - 9

Dimensionality Reduction using PCA

Breast Cancer Dataset

METHODOLOGY

Principal Component Analysis (PCA)

Linear feature extraction technique for high-dimensional data analysis.

DATASET

Breast Cancer Wisconsin Diagnostic Dataset

Multivariate medical diagnostic dataset used for feature reduction & classification.

OBJECTIVE

Reduce the number of features while retaining important information

Maximize variance retention while simplifying model complexity.

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What is Dimensionality Reduction?

DEFINITION

Variable Reduction

Dimensionality reduction reduces the number of input variables in a dataset.

CORE GOAL

Info Retention

It aims to preserve the most important information while using fewer dimensions.

ADVANTAGES

Key Benefits

Benefits include simpler visualization, reduced redundancy, and potentially lower computational cost.

ALGORITHMS

Techniques

Common techniques include PCA, Linear Discriminant Analysis (LDA), and t-SNE.

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Principal Component Analysis (PCA)

OVERVIEW

Feature Transformation

PCA transforms correlated features into a new set of uncorrelated variables called principal components.

PCA is an unsupervised dimensionality-reduction technique.

FIRST COMPONENT

Principal Component 1

Principal Component 1 captures the largest possible variance in the dataset.

ORTHOGONALITY

Principal Component 2

Principal Component 2 captures the next largest variance, subject to being orthogonal to PC1.

ORDERING

Explained Variance

Components are ordered by the amount of variance they explain.

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Breast Cancer Dataset

SAMPLE SIZE

569 Observations

Dataset contains 569 observations.

INPUT FEATURES

30 Diagnostic Features

There are 30 numerical diagnostic features after removing the ID column.

TARGET VARIABLE

Diagnosis Target

Target variable: diagnosis.

B = Benign; M = Malignant.

FEATURE MEASUREMENTS

Example Features

Includes radius_mean, texture_mean, perimeter_mean, area_mean, smoothness_mean, and related worst/standard-error measurements.

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Preprocessing Before PCA

STEP 01

Load Dataset

Load the CSV dataset into memory.

STEP 02

Drop ID

Remove the identifier column (id).

STEP 03

Separate Target

Separate diagnosis target from the 30 numerical features.

STEP 04

Handle Missing

Check and handle missing values in features.

STEP 05

Standardize

Standardize features using StandardScaler.

WHY STANDARDIZE?

PCA is scale-sensitive; features with larger numerical scales can otherwise dominate the analysis.

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Applying PCA

CONFIGURATION

Variance Target

PCA can be configured to retain a desired proportion of variance. In this experiment, **n_components = 0.95** was used.

SELECTION RULE

95% Retention

Automatically selects the smallest number of principal components that retain approximately **95% of the total variance** in the training data.

FEATURE REDUCTION

30 → 10 Components

The dataset was successfully reduced from **30 original features** down to just **10 principal components**.

KEY OUTCOME

A **66% reduction** in feature dimensions (from 30 to 10) was achieved while maintaining 95% of the dataset's total variance.

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Comparing Model Performance

To assess the practical effect of PCA, use the same classifier and the same train/test split.

BASELINE SETUP

Without PCA

30 Standardized Features

Train the Logistic Regression classifier using all 30 original standardized features to establish baseline predictive performance.

REDUCED DIMENSIONS

With PCA

10 Principal Components

Train the identical classifier using only the top 10 principal components to evaluate speed and accuracy trade-offs.

EVALUATION METRICS

Compare accuracy and other key performance metrics such as precision, recall, F1-score, and total training time.

Model Choice: Logistic Regression | **Dataset split:** Identical train/test split

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Experimental Results

BASELINE SETUP

Without PCA

96.49% accuracy

- **30 Features** used
- Training time ≈ **0.065 s**

PCA OPTIMIZED

With PCA

97.37% accuracy

- **10 Components** retained
- Training time ≈ **0.066 s**

KEY TAKEAWAYS & COMPARISON

- **Dimensionality Reduction:** 30 → 10 components (**66.67% reduction**)
- **Accuracy Difference:** Improved by approximately **+0.88 percentage points**
- *Training times are essentially similar for this relatively small dataset.*

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Interpretation of Results

Key findings and practical implications from the principal component analysis experiment.

DIMENSIONALITY & VARIANCE

~95% Variance Retained

PCA substantially reduced the number of dimensions while retaining approximately 95% of the variance.

MODEL ACCURACY

Slightly Higher Accuracy

For this dataset and Logistic Regression model, accuracy was slightly higher after PCA.

COMPUTATIONAL SPEED

Comparable Training Time

The similar training times are expected because the dataset is small and model training is already very fast.

PRACTICAL TAKEAWAY

Context-Dependent Utility

PCA should not be described as universally better; its usefulness depends on the dataset, model, and evaluation criteria.

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Python Workflow

End-to-end data processing, principal component analysis, and model evaluation pipeline.

STEP 01

Data Ingestion

Load dataset using pandas

STEP 02

Data Cleaning

Drop id and separate diagnosis target

STEP 03

Train/Test Split

Split into training and testing sets

STEP 04

Feature Scaling

Standardize using StandardScaler

STEP 05

Fit PCA

Apply PCA (n_components=0.95) on training data

STEP 06

Transform Test Data

Transform the test data using fitted PCA

STEP 07

Model Training

Train Logistic Regression classifier

STEP 08

Model Evaluation

Evaluate & compare performance with baseline

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Result & Conclusion

Summary of dimensionality reduction and model accuracy on the Breast Cancer dataset.

DIMENSIONALITY REDUCTION

30 → 10 Components

~95% variance retained

PCA was successfully applied to the Breast Cancer dataset, reducing 30 original features to 10 principal components.

CLASSIFICATION ACCURACY

Logistic Regression Comparison

97.37% with PCA

The Logistic Regression model achieved **96.49%** accuracy without PCA and **97.37%** accuracy with PCA.

KEY TAKEAWAY

Thus, for this experiment, PCA achieved substantial dimensionality reduction while maintaining comparable classification performance.